

Supplementary note: finite-grid preparation and initial-rate algebra for additive coagulation–fragmentation

Setting

This note accompanies the paper *Sampling-law separation and finite nonlinear corrections in additive coagulation–fragmentation* and uses its notation. Sizes lie in $E = (0, \infty)$; n_t is the finite nonnegative measure of particle number per unit volume, $N(t) = n_t(E)$ its count, and $M_1(t) = \int_E x n_t(dx)$ its mass, conserved at the value $m > 0$. Coagulation has a measurable symmetric kernel $K(x, y) \geq 0$, and fragmentation has a per-particle selection rate $S(x) \geq 0$ with an expected daughter measure b_x on $(0, x)$ of total count two and total mass x . The additive model of the main paper is $K(x, y) = \lambda(x + y)$ and $S \equiv \sigma$, with $b = \lambda m$. For a finitely supported preparation v , its count vector field is

$$\mathcal{R}(v) = \int S(x)v(dx) - \frac{1}{2} \iint K(x, y)v(dx)v(dy).$$

The count-neutral classification theorem of the main paper states that $\mathcal{R}(v) = 0$ for every nonnegative finitely supported v with $M_1(v) = m$ if and only if there is a nonnegative pointwise finite measurable function d with

$$K(x, y) = xd(y) + yd(x), \quad S(x) = md(x). \quad (*)$$

1 Preparation constraints and elementary initial-rate tomography

This note records supporting algebra for the observation limitations discussed in the main paper’s section on bulk observables. Quadratic response models on mixture simplices are classical; see, for example, Brown et al. [1]. The statements below are direct applications of quadratic algebra, not a separate inverse-problem method. They concern exact initial rates under specified preparations. Full-trajectory conclusions are stated only when a conserved quantity or a closed count equation justifies them.

1.1 Quadratic rates on a finite preparation shell

On a grid of q selected sizes, let $z \in \mathbb{R}^q$ be the vector of number concentrations, K a symmetric kernel matrix, and s the vector of single-particle count production rates. Write $\mathcal{R}(z) = s^T z - z^T K z / 2$. The entries need not be positive for the following algebraic statement; physical admissibility is an additional constraint.

Proposition 1.1 (Finite preparation-shell classification). *Let $C \in \mathbb{R}^{r \times q}$ have full row rank, let $h \in \mathbb{R}^r \setminus \{0\}$, and suppose $\{z > 0 : Cz = h\}$ is nonempty. Then $\mathcal{R}$ vanishes on this shell if and only if there is $A \in \mathbb{R}^{r \times q}$ such that*

$$K = C^T A + A^T C, \quad s = A^T h. \quad (1.1)$$

In particular $\text{rank } K \leq 2r$.

Proof. A polynomial vanishing on a nonempty relatively open subset of an affine space vanishes on the whole affine space. Put $P = \mathbf{C}^T(\mathbf{C}\mathbf{C}^T)^{-1}\mathbf{C}$ and $Q = I - P$. The Hessian restricted to $\ker \mathbf{C}$ vanishes, so polarization gives $Q\mathbf{K}Q = 0$. Define

$$\mathbf{A}_0 = (\mathbf{C}\mathbf{C}^T)^{-1}\mathbf{C}\mathbf{K}(I - P/2).$$

A direct expansion using $Q\mathbf{K}Q = 0$ gives $\mathbf{C}^T\mathbf{A}_0 + \mathbf{A}_0^T\mathbf{C} = \mathbf{K}$. On the affine space the residual is $(s - \mathbf{A}_0^T h)^T z$. Its vanishing implies $s - \mathbf{A}_0^T h = \mathbf{C}^T \gamma$ and $\gamma^T h = 0$: the residual annihilates $\ker \mathbf{C}$, and evaluation at one feasible z gives the second identity. The skew matrix

$$L = \frac{h\gamma^T - \gamma h^T}{h^T h}$$

satisfies $L^T h = \gamma$. Hence $\mathbf{A} = \mathbf{A}_0 + L\mathbf{C}$ keeps the representation of $\mathbf{K}$ and gives $\mathbf{A}^T h = s$. The converse follows by substitution, and the rank bound follows from (1.1). $\square$

The rank bound is necessary, not sufficient for invisibility. For simultaneous constraints on count and mass, the rows of $\mathbf{C}$ are $(1, \dots, 1)$ and the grid sizes, with $h = (c, m)^T$. A corresponding continuum sufficient family is

$$K(x, y) = a(x) + a(y) + xd(y) + yd(x), \quad S(x) = ca(x) + md(x). \quad (1.2)$$

For bounded nonnegative a, d , every mass-conserving solution with a valid count balance has

$$N' = (c - N) \int a \, dn_t + (m - M_1) \int d \, dn_t.$$

If $N(0) = c$ and $M_1 = m$, the locally integrable coefficient $\int a \, dn_t$ and the scalar linear equation give $N(t) = c$. This is a conditional statement about such solutions, not a new forward existence theorem. The finite-grid classification does not by itself prove a continuum converse under two constraints. In contrast, the count-neutral classification theorem of the main paper proves the complete continuum classification when only initial mass is fixed and count is allowed to vary.

1.2 Dilution, sources, and the exact algebraic ambiguity

For this subsection allow a size-independent count source J and a net single-particle count production rate $\beta(x)$; these are observation parameters, distinct from $b = \lambda m$ in the additive model. For a preparation $n_0 = cp$, where p is a finitely supported probability measure, suppose the observed initial count rate is

$$\mathbf{r}(c, p) = J + c \int \beta \, dp - \frac{c^2}{2} \iint K(x, y) p(\mathrm{d}x) p(\mathrm{d}y). \quad (1.3)$$

Assume symmetric, pointwise finite K , known positive concentrations, and unchanged J, β, K across preparations. Antisymmetric parts of a kernel never enter this observation, which explains the symmetry assumption.

Proposition 1.2 (Fixed-concentration gauge). *With known J , two models agree in (1.3) for all p at one fixed concentration c if and only if their differences satisfy*

$$\Delta K(x, y) = a(x) + a(y), \quad \Delta \beta(x) = ca(x) \quad (1.4)$$

for some function a . Agreement at two distinct positive concentrations removes this ambiguity. With unknown J , agreement at two distinct positive concentrations c_1, c_2 leaves exactly the algebraic family

$$\Delta J = \zeta, \quad \Delta \beta = -\zeta(c_1^{-1} + c_2^{-1}), \quad \Delta K = -\frac{2\zeta}{c_1 c_2}. \quad (1.5)$$

A third distinct concentration removes this family.

Proof. For known J , monodisperse preparations give $\Delta\beta(x) = c\Delta K(x, x)/2$. Equal mixtures of two atoms give $\Delta K(x, y) = [\Delta K(x, x) + \Delta K(y, y)]/2$. These are (1.4), and substitution proves sufficiency. Applying the diagonal identity at two concentrations forces both differences to vanish. For unknown J , monodisperse data at c_1, c_2 give the two linear equations $0 = \zeta + c_j\Delta\beta(x) - c_j^2\Delta K(x, x)/2$. Solving gives the constants in (1.5); equal mixtures then give the same constant off the diagonal. Their rate difference at any concentration c is $\zeta(1 - c/c_1)(1 - c/c_2)$, proving both remaining assertions. $\square$

Nonnegativity can restrict the admissible values of ζ or force $\zeta = 0$ at a boundary model. Thus the algebraic family does not assert physical nonuniqueness at every parameter value. Diluting the same shape p changes both count and material loading. It differs from changing p while preserving $\int x \, dn_0 = m$, which leaves the entire count-neutral class $(*)$ invisible.

1.3 Finite-grid reconstruction and noise bounds

For known J and any $c > 0$, direct substitution gives

$$K(x, y) = \frac{\mathfrak{r}(c, \delta_x) + \mathfrak{r}(c, \delta_y) - \mathfrak{r}(2c, (\delta_x + \delta_y)/2) - J}{c^2}, \quad (1.6)$$

$$\beta(x) = \frac{4\mathfrak{r}(c, \delta_x) - \mathfrak{r}(2c, \delta_x) - 3J}{2c}. \quad (1.7)$$

On q selected sizes this uses q monodisperse rates at c and $q(q+1)/2$ equal-mixture rates at $2c$, including repeated sizes. The total equals the number of unrestricted symmetric-kernel and linear-rate parameters. The observation map is linear in those parameters; fewer scalar initial-rate observations cannot be injective on an open parameter set. This is only a measurement-count bound. It makes no claim of optimal noise performance or optimal use of finite-time trajectories. If J is unknown, one further rate suffices, because for a reference size x ,

$$J = 3\mathfrak{r}(c, \delta_x) - 3\mathfrak{r}(2c, \delta_x) + \mathfrak{r}(3c, \delta_x). \quad (1.8)$$

With exact J and absolute errors at most ε in each rate, equations (1.6) and (1.7) imply

$$|\widehat{K} - K| \leq 3\varepsilon/c^2, \quad |\widehat{\beta} - \beta| \leq 5\varepsilon/(2c). \quad (1.9)$$

If J is estimated with error at most ε_J , the right sides increase by ε_J/c^2 and $3\varepsilon_J/(2c)$, respectively. Using (1.8) with the same rate-error bound gives $\varepsilon_J \leq 7\varepsilon$; correlations among reused observations do not affect these deterministic inequalities.

Suppose now the count difference $N(t) - N(0)$ has error at most η and each required trajectory has $|N''(s)| \leq L$ for $0 \leq s \leq t$. Taylor's integral remainder gives initial-rate error at most $\eta/t + Lt/2$. With exact J this yields

$$|\widehat{K} - K| \leq \frac{3}{c^2} \left(\frac{\eta}{t} + \frac{Lt}{2} \right). \quad (1.10)$$

When $\eta, L > 0$, the bound is minimized at $t = \sqrt{2\eta/L}$ if that time lies in the interval where the curvature bound is known; the resulting bound is $3\sqrt{2L\eta}/c^2$. Zero-error or zero-curvature cases are read directly from (1.10). An exact initial count and terminal count error at most η are one way to meet the assumed count-difference error bound; if both are noisy, their errors must be combined.

A uniform curvature bound is an additional assumption and can worsen with concentration, so these formulas do not justify arbitrarily large c . Finite-width preparations measure kernel averages; localization error requires a separate regularity bound. Dilution must preserve the kinetic coefficients for (1.3) to apply across experiments. Changes in the surrounding medium can violate

this assumption. Collision-induced fragmentation also contributes a quadratic term and cannot be separated from coagulation by concentration scaling alone. None of these count-rate observations identifies the daughter measure; the distributional data used for late-time Fourier identification in the main paper contain information that count measurements lack.

References

- [1] L. Brown, A. N. Donev, and A. C. Bissett. General blending models for data from mixture experiments. *Technometrics*, 57(4):449–456, 2015. doi: 10.1080/00401706.2014.947003. URL <https://pmc.ncbi.nlm.nih.gov/articles/PMC4673519/>.